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Printed Pages : 4 **EC – 607**

(Following Paper ID and Roll No. to be filled in your Answer Book)

PAPER ID : 3043 Roll No.

B. Tech.

(SEM. VI) EXAMINATION, 2006-07

COMMUNICATION ENGINEERING

Time : 3 Hours] *[Total Marks : 100*

Note : *Attempt **all** questions, All questions carry **equal** marks.*

1. Attempt any **four** parts of the following. **5×4=20**

(a) Evaluate the Fourier transform of the following signal **5**

$$g(t) = \exp(-t)^{*} \sin(2\pi f_c t)$$

where '*' denotes the convolution operator.

(b) Derive an expression for the effective modulation index of a multi-tone modulated AM signal. **5**

(c) Describe briefly the operation of a Ring modulator used for the generation of DSB-SC signals. **5**

(d) Describe briefly the operation of the superheterodyne receiver system. **5**

Source: AKTU official examination paper - www.aktunotes.in

(e) Consider the narrow-band FM signal approximately defined by **2+2+1**

$$s(t) = A_c \cos(2\pi f_c t) - \beta \sin(2\pi f_c t)\sin(2\pi f_m t)$$

Determine the envelope of this modulated signal. What is the ratio of the maximum to the minimum value of this envelope? Draw a phasor diagram to represent the above signal.

(f) Describe briefly the indirect method of generation of FM signals. **5**

2 Attempt any **four** parts of the following. **5×4=20**

(a) What do you mean by the narrow-band noise? Discuss briefly, the properties of the narrow-band noise. **5**

(b) Describe briefly the functions of pre-emphasis and de-emphasis filters in FM systems. **5**

(c) State and explain the sampling theorem for band-pass signals. **5**

(d) Explain natural and flat-top sampling. Compare the two. **5**

(e) Describe briefly the Pulse Width Modulation technique. **5**

(f) Explain the methods for the demodulation of PAM signals. **5**

3 Attempt any **two** parts of the following. **2×10**

(a) What is quantization noise in PCM systems? How does it depend upon the step size? Suggest some methods to overcome the difficulties encountered when the modulating signal amplitude swing is large. **2+3+5**

(b) Describe delta modulation systems. What are its limitations? How can they be overcome? **5+2+3**

(c) Describe briefly the different line codes used for the electrical representation of the binary data 0 and 1. **10**

4 Attempt any **two** parts of the following. **2×10**

(a) What are TDM and FDM systems? Compare the two and mention the application of each. **10**

(b) Discuss briefly the operation of a noncoherent FSK transmitter and receiver systems. Also derive an expression for the probability of error. **3+7**

(c) Explain phase shift keying. Describe coherent detection of PSK signals. Also derive an expression for the probability of error. **10**

5 Attempt any **two** parts of the following: **2×10**

(a) What do you mean by the mutual information of a communication channel? Define the capacity of a discrete memoryless channel in terms of mutual information. Derive an expression for the channel capacity of a discrete memoryless binary symmetric channel. **3+2+5**

(b) Consider a sequence of symbols emitted by a source with their probabilities as given below: **8+2**

Symbol	**x1**	**x2**	**x3**	**x4**	**x5**	**x6**	**x7**	**x8**
Probability	**0.13**	**0.07**	**0.19**	**0.11**	**0.1**	**0.15**	**0.15**	**0.1**

Compute the Huffman code for the above source symbols. Also find the average code-word length.

(c) Consider a sequence of symbols generated by a source with their probabilities of occurrence as given below: **8+2**

Symbol	**s1**	**s2**	**s3**	**s4**	**s5**	**s6**	**s7**	**s8**	**s9**
Probability	**0.22**	**0.19**	**0.15**	**0.12**	**0.08**	**0.06**	**0.06**	**0.07**	**0.05**

Determine the code words of the symbols using Shannon-Fano coding technique. Also determine the average code-word length.
